ARTIFICIAL INTELLIGENCE

Lab Report

LAB TASK 2

Self-Reflex Agent: Weather Prediction System

SUBMITTED BY

Maira Jamil

STUDENT ID

41362

SUBMITTED TO

Mam Faiza Ali

COURSE

Artificial Intelligence (Lab)

Task Statement

Apply your knowledge of AI Agents to the Self Reflex agent exercise and share the Colab file as a PDF with justification where required.

Notebook link: https://www.kaggle.com/code/scratchpad/notebook744432ad00/edit

Implementation and Output

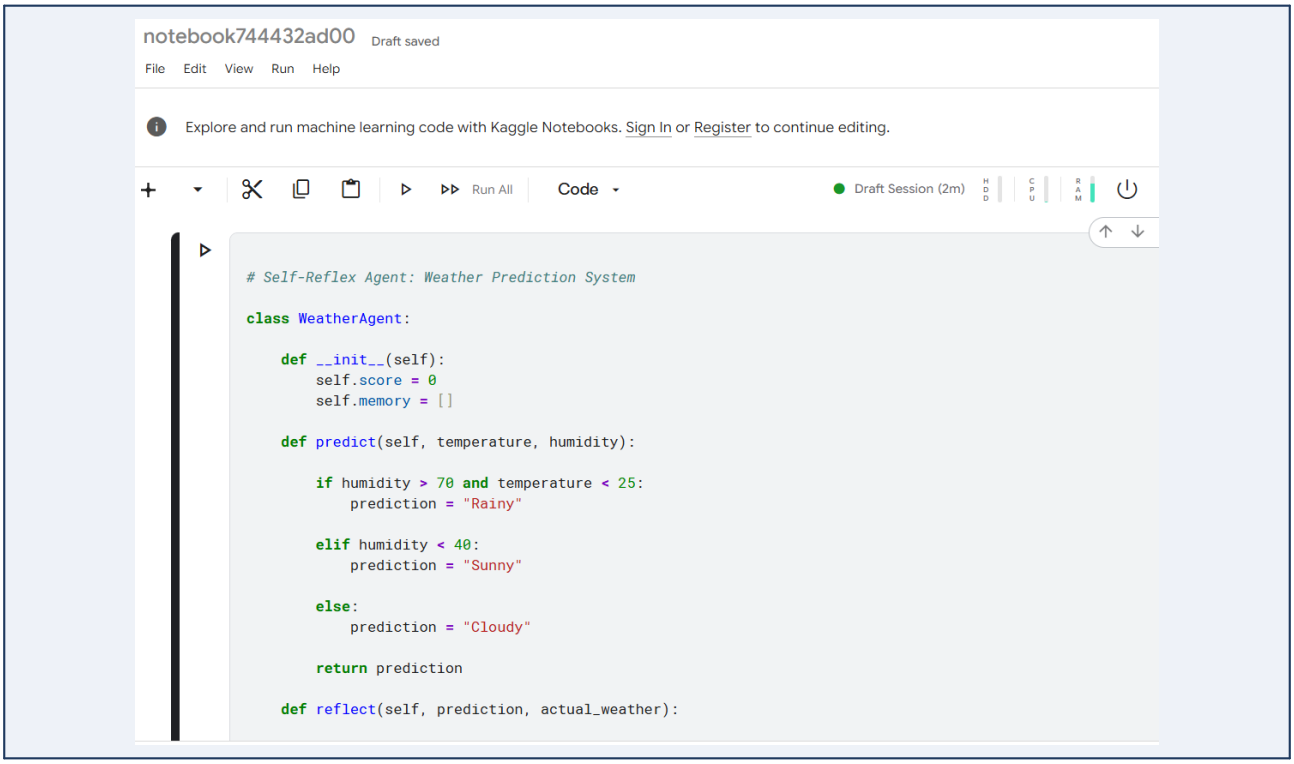
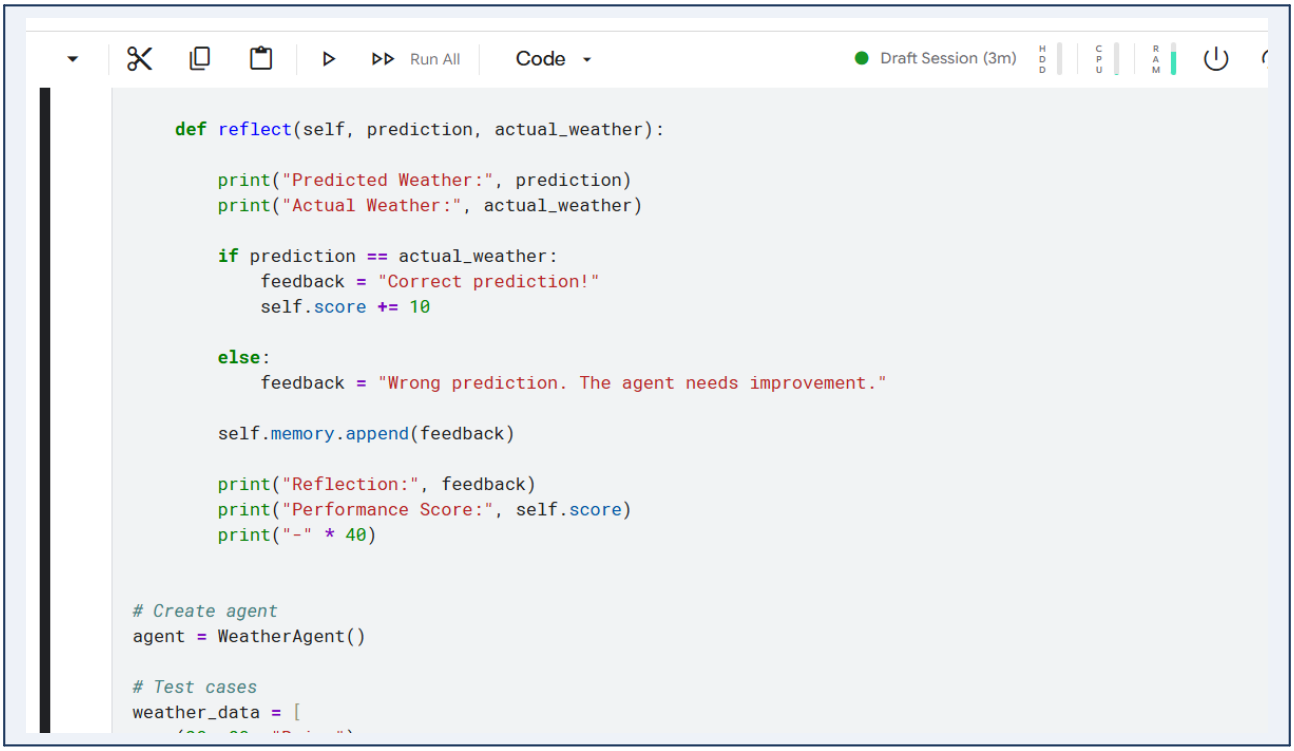


Figure 1: WeatherAgent class and predict() method



```
    def reflect(self, prediction, actual_weather):

        print("Predicted Weather:", prediction)
        print("Actual Weather:", actual_weather)

        if prediction == actual_weather:
            feedback = "Correct prediction!"
            self.score += 10

        else:
            feedback = "Wrong prediction. The agent needs improvement."

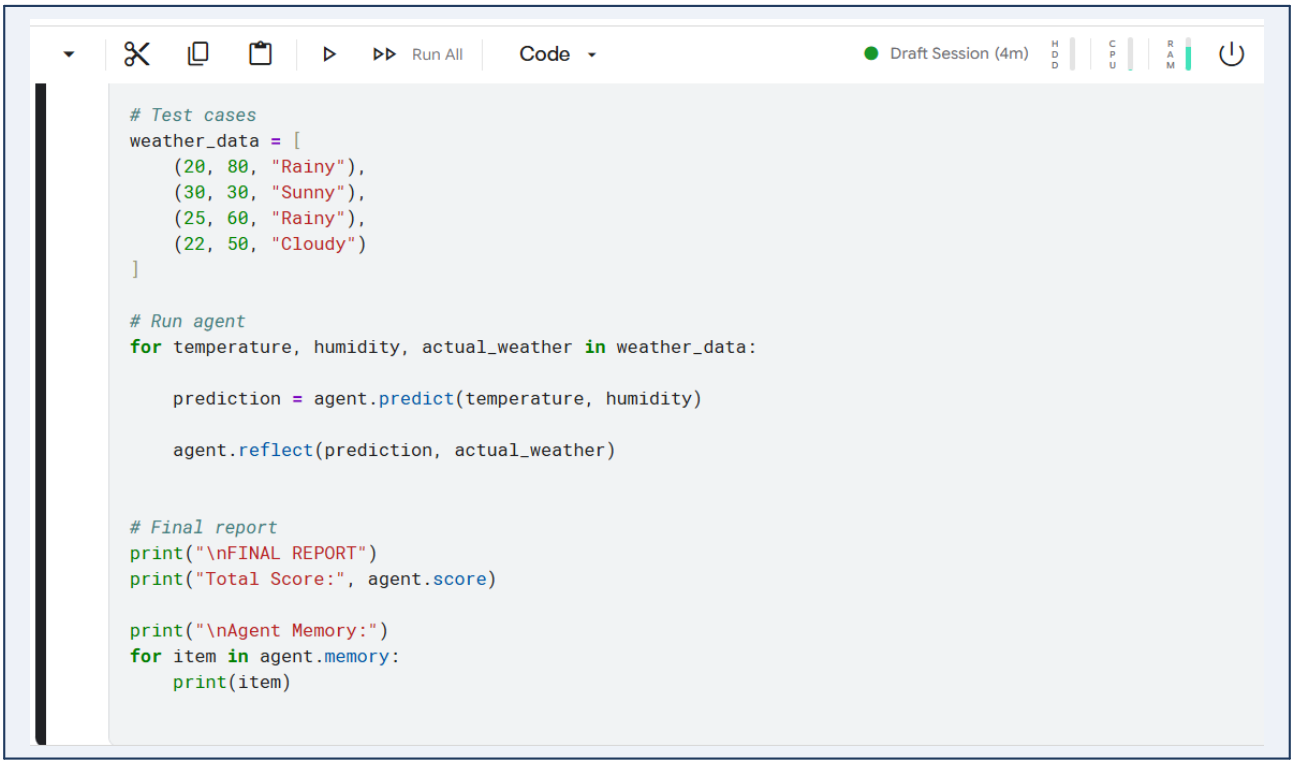
        self.memory.append(feedback)

        print("Reflection:", feedback)
        print("Performance Score:", self.score)
        print("-" * 40)


# Create agent
agent = WeatherAgent()

# Test cases
weather_data = [
```

Figure 2: reflect() method and agent creation



```
# Test cases
weather_data = [
    (20, 80, "Rainy"),
    (30, 30, "Sunny"),
    (25, 60, "Rainy"),
    (22, 50, "Cloudy")
]

# Run agent
for temperature, humidity, actual_weather in weather_data:

    prediction = agent.predict(temperature, humidity)

    agent.reflect(prediction, actual_weather)


# Final report
print("\nFINAL REPORT")
print("Total Score:", agent.score)

print("\nAgent Memory:")
for item in agent.memory:
    print(item)
```

Figure 3: Test cases, agent run loop and final report

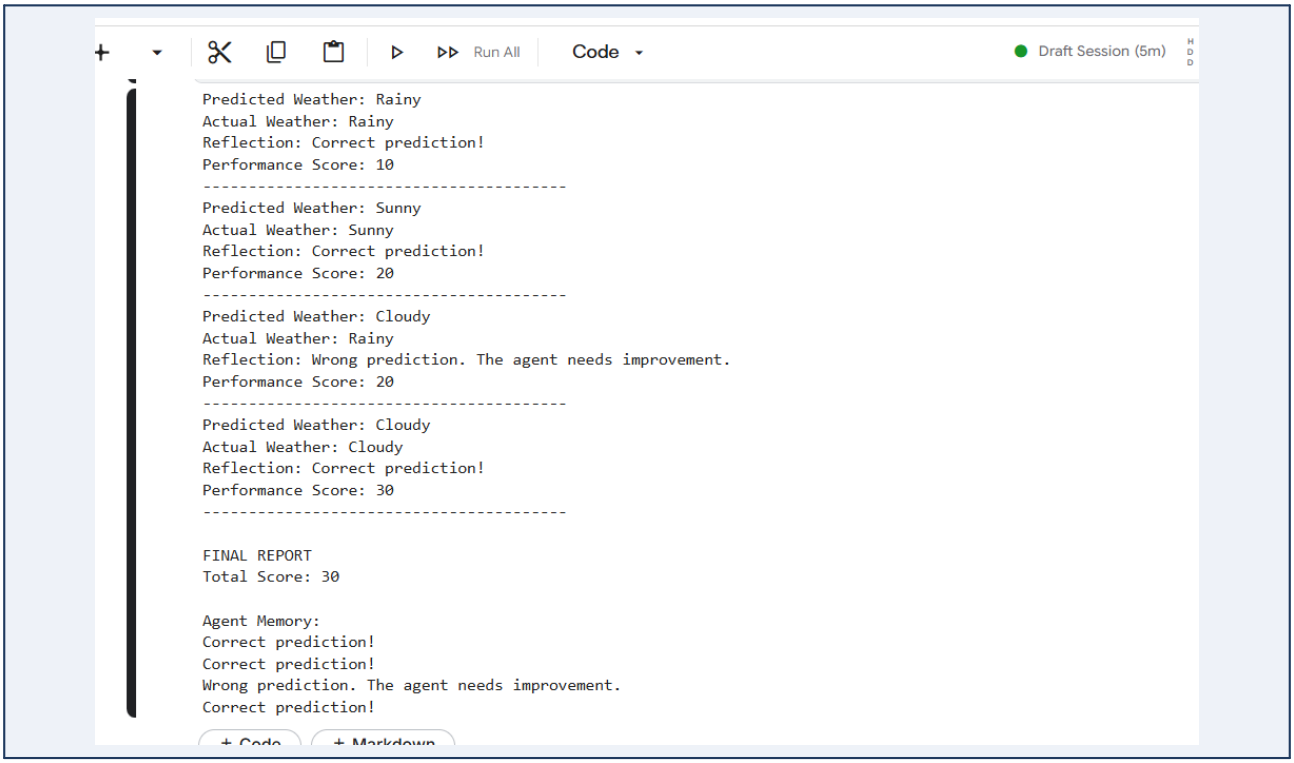


Figure 4: Program output

Justification

Why this is a self-reflex agent. The agent's *predict()* method selects its output from the current percept (temperature and humidity) using fixed condition-action rules, without planning ahead or modelling the world. The *reflect()* method then compares the prediction with the actual weather, awards 10 points for a correct prediction, and stores feedback in memory.

Rules and results

Temp	Humidity	Rule applied	Predicted	Actual	Result
20	80	humidity > 70 and temp < 25	Rainy	Rainy	Correct
30	30	humidity < 40	Sunny	Sunny	Correct
25	60	else (temp is not below 25)	Cloudy	Rainy	Wrong
22	50	else (no rule matched)	Cloudy	Cloudy	Correct

Final score: 30. Three of the four predictions were correct (3 x 10 = 30). The third case (25, 60) was wrong because the rain rule requires a temperature strictly below 25, so the agent fell through to the default "Cloudy". This shows the main limitation of a simple reflex agent: it can only be as good as its hand-written rules, and the stored memory records the mistake but does not change future behaviour.